| Roll No (TE-CMPN-A) | Name |
|---|---|
| 25102A2001 | Siddhi Kharangate |
| 25102A2003 | Abhay Sonone |
| 25102A2004 | Sumit Shidole |
| 25102A2009 | Siya Dalal |

# Title: Changes Made to the Proposed Electrochemical Selective Recovery Process

## 1. Overview of Changes

The original process consisted of E-Waste Components → Pre-processing → Metal-Rich Solution → Electrochemical Separation → Selective Metal Deposition → Pure Strategic Metals. The process has been improved by adding intermediate preparation and post-recovery stages so that the proposed method presents a clearer and more complete technical workflow.

## 2. Change 1: E-Waste Collection & Sorting

The first stage has been expanded from only "E-Waste Components" to "E-Waste Collection & Sorting". In this stage, electronic waste is collected and sorted to identify and separate metal-containing components from unwanted non-metallic materials. This improves the input quality before further processing and makes the recovery process more organized.

## 3. Change 2: Dismantling & Pre-processing

Pre-processing is now described in greater detail. The selected electronic components are dismantled and mechanically processed to expose and concentrate the metal-containing material. This creates a more suitable feed material for preparing the metal-rich solution.

## 4. Change 3: Metal-Rich Solution Preparation

The original "Metal-Rich Solution" stage has been expanded to explain how the processed e-waste is treated so that valuable metals are converted into dissolved metal ions. The resulting solution becomes the feed for the electrochemical recovery stage.

## 5. Change 4: Filtration & Electrolyte Preparation

A new intermediate stage has been added before electrochemical recovery. The metal-rich solution is filtered to remove unwanted solid particles and is prepared as a suitable electrolyte. This provides a cleaner and more appropriate solution for controlled electrochemical treatment.

## 6. Change 5: Controlled Electrochemical Separation & Selective Deposition

The former electrochemical separation and selective deposition stages are now described together as a controlled recovery stage while retaining both concepts. A controlled electrical potential is applied between the electrodes. Metal ions move under the electric field and are reduced at the cathode, where they form solid metal deposits. The process can be controlled using applied potential, current density, electrolyte composition and electrode material.

## 7. Change 6: Metal Collection, Purification & Reuse

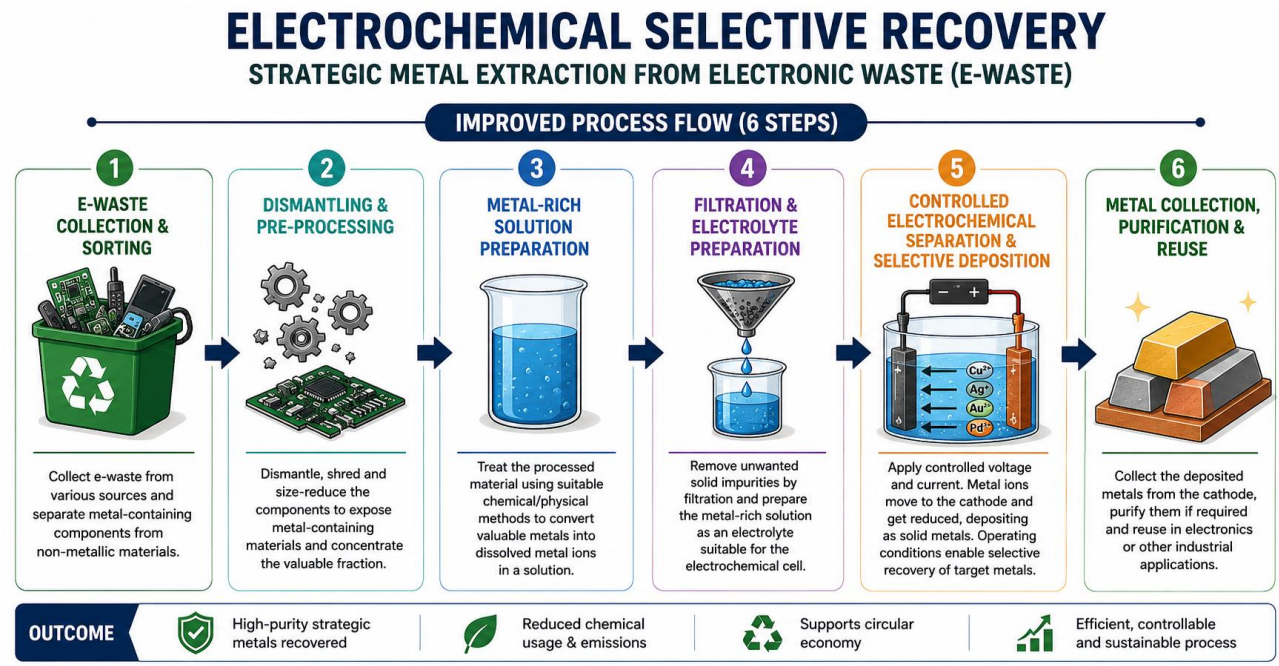
The final stage has been expanded from “Pure Strategic Metals” to “Metal Collection, Purification & Reuse”. After deposition, the metal is removed from the cathode and purified if required. The recovered metal can then be reused in electronic products or other industrial applications.

## 8. Improved Process Flow

E-Waste Collection & Sorting → Dismantling & Pre-processing → Metal-Rich Solution Preparation → Filtration & Electrolyte Preparation → Controlled Electrochemical Separation & Selective Deposition → Metal Collection, Purification & Reuse

## 9. Main Improvement

The main improvement is that the process now includes preparation of the e-waste before solution formation, conditioning of the metal-rich solution before electrochemical treatment, controlled operating conditions during deposition, and explicit collection, purification and reuse after recovery. This makes the proposed process more complete and clearly explains how the recovered strategic metals move from e-waste to a reusable form.



# ELECTROCHEMICAL SELECTIVE RECOVERY

## STRATEGIC METAL EXTRACTION FROM ELECTRONIC WASTE (E-WASTE)

### IMPROVED PROCESS FLOW (6 STEPS)

**1 E-WASTE COLLECTION, & SORTING**



Collect e-waste from various sources and sort metal-containing components from non-metallic materials.

**2 DISMANTLING & PRE-PROCESSING**



Dismantle, shred and size-reduce the components to expose and concentrate metal-containing materials.

**3 METAL-RICH SOLUTION PREPARATION**



Treat the processed material to convert valuable metals into dissolved metal ions in a solution.

**4 FILTRATION & ELECTROLYTE PREPARATION**



Remove solid impurities by filtration and prepare the metal-rich solution as a suitable electrolyte.

**5 CONTROLLED ELECTROCHEMICAL SEPARATION & SELECTIVE DEPOSITION**



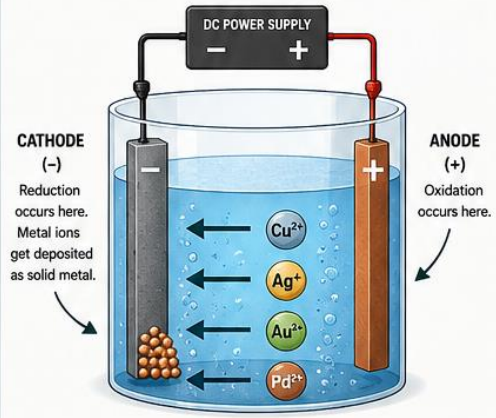
Apply controlled voltage and current. Metal ions move to the cathode and get reduced to deposit as solid metals.

**6 METAL COLLECTION, PURIFICATION & REUSE**



Collect the deposited metals, purify if required and reuse in electronics or other industrial applications.

### ELECTROCHEMICAL RECOVERY CELL




Example (Copper deposition):

$Cu^{2+} + 2e^- \longrightarrow Cu\ (s)$

### KEY PROCESS PARAMETERS

- **Applied Voltage / Potential** — Controls the electrochemical reaction and determines which metal ions are deposited.
- **Current Density** — Influences the rate and quality of metal deposition.
- **Electrolyte Composition** — Affects metal-ion movement, selectivity and recovery efficiency.
- **Electrode Material** — Influences efficiency, stability and purity of recovered metals.
- **Solution Conditions (pH, Temperature)** — Affect ion availability, conductivity and deposition quality.

### WORKING PRINCIPLE

1. Metal ions are present in the electrolyte solution.
2. Under an electric field, ions move towards the cathode.
3. Ions gain electrons to get reduced.
4. Reduced ions form solid metal on the cathode surface.
5. Deposited metal is collected, purified and reused.

### WHAT MAKES THIS METHOD NOVEL?

This method uses controlled electrical energy and adjustable operating parameters (voltage, current density, electrolyte composition, electrode material) to selectively recover specific strategic metals from e-waste, providing a cleaner, more efficient and sustainable alternative to conventional recovery techniques.

**Supports Circular Economy**

### ADVANTAGES

- **Selective Recovery** — Adjust conditions to target specific metal ions.
- **High-Purity Deposition** — Metals are deposited as solid layers with high purity.
- **Reduced Chemical Dependence** — Lower use of harsh chemicals compared to conventional methods.
- **Better Process Control** — Voltage, current and other parameters can be precisely monitored.
- **Resource Recovery** — Valuable metals are returned to the manufacturing cycle.
- **Environment Friendly** — Lower environmental impact and supports sustainable development.

### APPLICATIONS

- Recovery of Copper from printed circuit boards.
- Recovery of Silver, Gold & Palladium from electronic components.
- Treatment of metal-rich solutions generated during e-waste recycling.
- Recovery and purification of strategic metals for reuse in electronics and other industries.

### OVERALL BENEFITS

